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Smart Home Automation Using Arduino

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Electrical and Electronics Engineering

by

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May 2026

DECLARATION

We here by declare that the thesis entitled "**Smart Home Automation Using Arduino**" submitted by **Paras Singh Bisht, Snehil Raj** for the award of the degree of *Bachelor of Technology in Electrical and Electronics Engineering* to VIT University is a record of bonafide work carried out by us under the supervision of **Dr. Sathish Kumar K**, Professor Grade 2, School Of Electrical Engineering, VIT University, Vellore.

We further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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Executive Summary

With the growing need for energy-saving and smart living spaces, smart home automation systems are gaining popularity. In this thesis, we propose a design and implementation of a Smart Home Automation System based on Arduino and Internet of Things (IoT). This system is designed to automatically control home appliances based on environmental factors, and also offers user control and remote operation.

The system employs a dual controller design, with separate controllers for sensing and decision-making, and output and user interface. The environmental conditions are noted using sensors such as Light Dependent Resistor (LDR), infrared (IR) sensors and a temperature sensor. These are then used to automatically control lighting and fans. The system incorporates a Wi-Fi module for remote monitoring via a mobile app and uses a keypad for security lock and secure the door.

The design exhibits energy savings, increased user comfort and robust system operation. The system showcases the potential of integrating embedded systems and IoT for automation in the modern era and offers a blueprint for future smart homes.

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LIST OF TERMS AND ABBREVIATIONS

IoT Internet of Things

LDR Light Dependent Resistor

IR Infrared

PWM Pulse Width Modulation

UART Universal Asynchronous Receiver Transmitter

I2C Inter-Integrated Circuit

LCD Liquid Crystal Display

IDE Integrated Development Environment

WiFi Wireless Fidelity

API Application Programming Interface

DC Direct Current

AC Alternating Current

TCPIP Transmission Control Protocol / Internet Protocol

CHAPTER 1

INTRODUCTION

1.1 Objective

The aim of the project is to develop a Smart Home Appliance Automation System based on Arduino and IoT (Internet of Things) to provide effective monitoring and control of appliances.

The system seeks to control appliances like lights, fans and security devices automatically using sensor feedback and control algorithms. It combines various technologies such as microcontrollers, sensors, and wireless communication technologies to deliver a robust and smart automation system.

One of the project aim is to create a system that employs two microcontrollers (Arduinos), with one dedicated to sensor data evaluation and decision-making, while the other is used for output control and communication with the user. This will improve the overall speed and efficiency of the system.

Another aspect of the project is to facilitate remote access via IoT, enabling appliance control through a mobile app. A keypad lock is also added for security and to prevent unauthorised access.

The aim of the project are to:

1. To develop a system to automate home appliances.
2. To provide sensor-based control based on ambient light, temperature and occupancy.
3. To allow remote monitoring and control via IoT.
4. To design a locking system with a keypad.
5. To enable energy efficiency with smart automation.

1.2 Motivation

With the growing need for energy efficient and smart systems, home automation technologies have evolved to provide convenience and reduce human effort. In conventional systems, appliances are controlled manually, which may be inefficient and cumbersome in today's automated world.

The need for convenience and comfort in performing day-to-day tasks is the driving force behind this project. The evolution of the Internet of Things (IoT) has made it feasible to connect devices and manage them remotely through mobile devices and other online platforms (Alheraish, 2016). This enables remote control of appliances, offering flexibility and convenience.

Energy saving is another driving factor. Devices are frequently left unused, which can lead to higher energy use. Sensors like the Light Dependent Resistor (LDR), Infrared (IR), and temperature sensors can be used to automatically switch appliances on and off according to different lighting and temperature conditions, thus avoiding wastage of energy (Pavithra and Balakrishnan, 2015).

Another factor driving the project is security. The keypad-based security system provides access control, improving safety and security.

Additionally, existing home automation systems are often expensive and complicated. This project will design an affordable and user-friendly system with Arduino, bringing automation to the masses (Arduino, 2020).

The motivational statement about the project should be what has done previously related to this project, that to make you want to do the project that you are interested in, what you hope to get out of the project, and what you want to apply new thing in the project.

1.3 Background

Home automation systems are used to automate and manage household appliances electronically and through communication networks. These systems involve the use of sensors, microcontrollers and actuators to develop an intelligent system that reacts to user commands and environmental changes.

Previous home automation systems used wired communication that was inflexible and difficult to implement. The development of wireless communication standards, including Bluetooth, Wi-Fi (Wireless Fidelity) and Global System for Mobile Communication (GSM), has made today's systems more effective and practical to deploy (Sriskanthan and Tan, 2002).

The advent of the Internet of Things (IoT) has enhanced home automation by facilitating communication between devices via the internet. IoT systems enable remote monitoring and control of appliances and enable automation based on environmental conditions (Alheraish, 2016).

Arduino microcontrollers are commonly employed in home automation systems due to their ease of use, affordability and versatility. They can be readily integrated with sensors and communication technologies, making them ideal for designing personalised automation systems (Arduino, 2020).

The aim of this project is to design a smart home automation system using a two-Arduino system and IoT. It involves sensors for data acquisition, microcontrollers for data manipulation, and wireless communication for remote control. This combination offers a smart, scalable and easy-to-use automation solution that increases ease of use, energy efficiency and control of appliances.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

2.1 Review of Literature

The development of home automation systems has evolved. The initial systems mainly relied on wired communication, which involved significant installation work and was not flexible. They were not easily reconfigurable, so couldn't adapt to today's changing lifestyles.

The emergence of wireless communication technologies has led to the exploration of Bluetooth, Global System for Mobile (GSM), ZigBee and Wireless Fidelity (Wi-Fi) technologies. Bluetooth-based systems are easy to implement and have low power consumption but are short range (Sriskanthan and Tan, 2002). GSM-based systems can be remotely controlled but are slow and expensive to operate.

ZigBee-based systems have low power consumption and support mesh network, which is suitable for large-scale automation. But they need more configuration and devices, adding complexity. In contrast, Wi-Fi systems offer faster data rates and easy integration with web-based applications, making them more appealing for smart home automation.

The emergence of the Internet of Things (IoT) has revolutionised home automation systems by providing real-time communication among devices using the internet. IoT systems enable remote monitoring and control of appliances through mobile devices or web applications (Alheraish, 2016). It also facilitates automation in response to environmental factors and user preferences.

Microcontroller platforms like Arduino are commonly used in smart home automation systems for their cost, versatility and ease of use. Arduino can be connected with various sensors and communication devices, making it suitable for building custom automation solutions (Arduino, 2020).

The latest trends in smart home systems include enhancing energy efficiency and automation. These systems can automatically switch appliances on and off with the help of sensors such as Light Dependent Resistor (LDR), infrared (IR) sensors and temperature sensors, which can help save energy (Pavithra and Balakrishnan, 2015).

However, current systems suffer from other problems such as high cost, difficult to install, not scalable, and lack of user-friendly interfaces (Chan et al., 2008). IoT systems also pose security challenges.

Thus, there is a requirement for a low-cost, scalable, secure and user-friendly system. The proposed system overcomes these challenges by integrating Arduino-based control with IoT and sensor-based automation.

2.2 Project Description

This project is a Smart Home Automation System using Arduino and Internet of Things (IoT) that offers intelligent control of appliances through automatic, manual and remote control.

It is designed using a distributed architecture with two microcontrollers (Arduino boards):

- Master Unit (Arduino 1): Handles data acquisition from sensors, information processing and decision-making based on logic rules.
- Slave Unit (Arduino 2): Responsible for output device control and user interaction.

This modular approach enhances system performance by distributing tasks across the controllers, sharing the processing burden, and providing fault tolerance.

The system uses various sensors for environmental monitoring:

- LDR for light detection.
- IR sensors for occupancy detection.
- DHT11 sensor for temperature measurement.

The system intelligently operates devices like lights and fans based on sensor data. For instance, lights are automatically switched on when it's dark and someone is home, and fans are regulated based on temperature.

The system is connected to the Internet of Things (IoT) platform for wireless communication using the ESP8266 module. This allows remote control of electrical appliances via a smartphone.

The system also features a keypad-based security system, enabling access control for the door. Also, the system has a feedback LCD screen.

The project has three modes of operation:

- Automatic Mode: Based on sensor inputs.
- Manual Mode: Using keypad input.
- Remote Mode: Using mobile application.

This alternate operation allows flexibility and enhanced convenience (Gill et al., 2009).

2.2.1 Software

The software is responsible for performing operations, data processing and inter-module communication.

The system is designed using Arduino Integrated Development Environment (IDE) and programmed in Embedded C. The software is developed using modular architecture, in which various operations are divided into modules.

The main software modules include:

- Data Acquisition Module: Acquires data from LDR, IR and temperature sensors.
- Control Module: Implements control logic to decide system actions.
- Communication Module: Manages serial communication between Arduino and ESP8266.
- User Interface Module: Handles input from keyboard and app.

The control logic runs in a loop - it continuously monitors and controls the system. It also has priority management, allowing manual control by the user to override automatic control mechanisms.

This design improves system robustness, ease of troubleshooting, and provides flexibility for future system upgrades (Arduino, 2021).

2.2.2 Hardware

Hardware refers to the physical components of the system. This encompasses microcontrollers, sensors, actuators, communication devices and power sources.

The microcontroller (Arduino Uno) is the main processing unit that runs the control program and manages the system.

The ESP8266 communication module offers wireless connectivity, allowing it to connect to the IoT platform and mobile app.

Sensors are used to collect environmental data:

- LDR for light intensity detection.
- IR sensors for occupancy detection.
- DHT11 for temperature measurement.

Output devices include:

- LED for lighting control.
- L298N motor driver for fan speed control using Pulse Width Modulation (PWM).
- Servo motor for door locking mechanism.
- Buzzer for alert signals.
- LCD screen for monitoring.

The system is properly grounded and regulated to ensure safe and consistent operation. The devices are connected with common ground to prevent any signal problems (Espressif Systems, 2020).

2.3 Goals

This project aims to create a smart home automation system that promotes convenience, efficiency and security in homes. The project goals are:

- To remotely control appliances using sensors.
- To allow remote access and monitoring through Internet of Things (IoT).
- To conserve energy by automation.
- To enhance security with keypad lock system.
- To create a low-cost and scalable solution.
- To provide an option for multi-mode operation

Home automation systems play an important role in today's world by improving resource management, comfort and convenience (Pavithra and Balakrishnan, 2015).

The project also intends to offer a hands-on experience with embedded systems, wireless communication and system integration.

CHAPTER 3

TECHNICAL SPECIFICATION

3.1 Software Specification

The software part of the Smart Home Automation System controls the operation of the entire system, manages the data obtained from sensors, communicates between different components of the system and provides real-time responses to changes in the environment and user requests.

The system is built with the Arduino Integrated Development Environment (IDE) and written in Embedded C to have direct control over the hardware and to achieve efficient execution of control algorithms.

- **Software Environment**

The development environment includes:

- Arduino Integrated Development Environment (IDE) for coding, compilation and program upload.
- Serial Monitor for debugging and testing.
- Libraries for sensor interface and communication.

This approach makes development easier with the use of Arduino IDE and enables embedded applications.

- **Programming Approach**

The software is programmed in the modular approach, with different features being implemented as modules. Main modules include:

- Sensor data acquisition.
- Decision-making logic.
- Communication handling.
- Output control.
- User interface management.

- **Sensor Data Processing**

The software takes input from sensors:

- LDR for light intensity.
- IR sensors for occupancy.
- DHT11 for temperature.

Threshold-based logic is used to control the system based on the data.

- **Control Logic Implementation**

The control logic is implemented with if-else statements and loops. The system is real-time as it constantly assesses sensor readings and produces the corresponding outputs.

Examples: Automatic switch of lights according to darkness and occupancy Fan speed based on temperature.

- **Communication Handling**

Communication is done via serial:

- SoftwareSerial: Communication between Arduino 1 and Arduino 2.
- UART: Communication between Arduino and ESP8266.
Data is sent in a mini format like:
- L1 / L0 for lighting.
- F0–F3 for fan speed.
- D1 / D0 for door control.

This helps in saving data and enhances efficiency.

- **IoT Integration**

The ESP8266 module is used to integrate with a mobile app. The software:

- Receives remote commands.
- Processes user inputs.
- Overrides automatic operation when required.

This enables flexibility/user interaction.

- **Software Reliability Features**

To enhance the system operation, the software has:

- Continuous looping for real-time operation.
- Input validation to avoid errors.
- Simple error handling mechanisms.
- Efficient memory usage.

3.2 Hardware Specification

The hardware system is the combination of all physical components, implemented with a distributed architecture using two Arduino boards to optimize performance.

- **Microcontroller Unit:** Arduino Uno boards with the ATmega328P microcontroller are utilized for the core logic.
 - **Characteristics:** Operates at 5V with a 16 MHz clock frequency and 32 KB flash memory.
 - **Configuration:** Includes a Master board for calculations and a Slave unit for execution.
- **Sensor Components:** Used to collect environmental data for system automation.
 - **LDR:** Measures light levels for automatic lighting control.
 - **IR Sensors:** Detect occupancy for automated presence-based actions.
 - **DHT11:** Monitors temperature levels for fan speed regulation.
- **Actuator Components:** Physical devices that perform actions based on sensor input.
 - **LCD:** Provide visual representation of the lighting system and real-time status feedback.
 - **Motors:** Includes an L298N driver for fan control and a Servo motor for the door locking mechanism.
 - **Buzzer:** Integrated to provide audible alert signals for security.

- **Communication and Power Systems:**
 - **ESP8266 Module:** Enables Wi-Fi connectivity and IoT functionality at 3.3V.
 - **Interconnection:** Uses serial connections (Arduino 1 ↔ Arduino 2 and Arduino ↔ ESP8266).
 - **Power Management:** Employs a strategy where high-power devices use external supplies to ensure stability.
- **Reliability Considerations:** Ensuring stable performance through proper grounding and minimized signal interference.

CHAPTER 4

DESIGN APPROACH AND DETAILS

4.1 Design Approach / Materials and Methods

The Smart Home Automation System design is motivated by the desire to build not just a functioning system, but also one that is resilient, scalable and user-friendly. The design focus is on distinct responsibilities, predictable data transfers, and robust operation under realistic conditions such as unstable power supply and sensor values, as well as intermittent network connectivity.

The design methodology involves hardware-software co-design, in which wiring, signal and voltage levels are considered at the same time the algorithm is designed. This allows the system hardware and control algorithms to work together, providing stable operation.

4.1.1 Design Methodology

- **Requirement Analysis**

The system requirements are mapped to behaviors:

- Automation: Light and fan automatically adjusts to ambient and occupancy.
- Responsiveness: Short time delay between input and action.
- Security: Limited access to door with failure feedback.
- Accessibility: Mobile remote control.
- Energy efficiency: Prevent needless load operation.

The design decisions are based on these requirements.

- **System Structuring**

The system is broken down into four blocks:

- Sensing (environment capture).
- Decision-making (logic evaluation).
- Execution (actuation).
- Communication (local + remote interaction).

This enables independent design, testing and optimisation of each block.

- **Dual Controller Strategy**

The use of two controllers is key to the design:

- Master polls sensors and evaluates rules.
- The slave is responsible for deterministic actuation and processing user input.

Benefits include: Reduced timing conflicts, Improved throughput, Simpler troubleshooting (isolate fault between sensor and actuation).

- **Mode Design (Automatic / Manual / Remote)**

The system has three modes of operation prioritised as follows:

- Automatic (sensor) - default
- Remote (IoT) - remote, user override
- Manual (keypad) - over-ride, local command
- Accessibility: Mobile remote control.
- Energy efficiency: Prevent needless load operation.

This gives priority to the user's intent.

4.1.2 System Design

The system is a closed loop, feedback control system.

- **Input Layer (Sensing)**

Sensors capture:

- Light level → determines lighting need.
- Occupancy → determines if control action is required.
- Temperature → determines cooling requirement.

Inputs are sampled often enough to represent the actual state of the system but not so often as to overwhelm the controller

- **Decision Layer (Logic Processing)**

The master controller processes inputs based on rules:

- Logs multiple switches (e.g., light + occupancy).
- Prevents chatter (e.g., due to small temperature variations).
- Creates tokens to be transmitted efficiently.

This layer is designed for deterministic operation - the same inputs will produce the same outputs.

- **Execution Layer (Actuation)**

The slave controller converts commands into actions:

- Digital outputs for ON/OFF devices.
- PWM signals for variable speed control.
- Timed signals for servo positioning.

Separation ensures that actuation is not blocked by sensor reading. beginitemize

- **Interaction Layer (User + Feedback)**

The system has three parts as follows:

- Keypad: secure local control.
- LCD: constant feedback (temperature, count, modes).
- Mobile App: remote control and monitoring.

This improves user-friendliness and transparency.

4.1.3 Interconnection design

Interconnections are designed for signal integrity, safety and clarity of data flow.

- **Common Ground Strategy**

Modules are connected to a common ground to:

- Prevent floating voltages
- Maintain uniform logic levels
- Prevent spurious operation due to VREF mismatch

- **Signal Flow Architecture**

A logical one way flow is maintained: Sensors → Master → Slave → Actuators
This eliminates feedback confusion and aids in diagnosis.

- **Serial Communication Design**

- Controller to controller: lightweight serial protocol on dedicated pins.
- Controller to Wi-Fi: UART.

Design considerations:

Short commands (e.g., L1, F2), Clear command semantics, Asynchronous to prevent blocking,

- **Voltage Level Considerations**

The Wi-Fi module logic level is different to the controllers. A voltage level adapter (divider) is used to:

- Protect the module.
- Ensure reliable data interpretation.

4.1.4 Connection Design

Connections grouped by function rather than pin number:

- Inputs → Master controller for centralised control.
- Outputs → Slave controller for efficient actuation.
- Communication links → Special pins for isolation.

Table 4.1 Arduino 1 (Master Unit) Connections

Component	Pin	Arduino 1 Connection
LDR Module	DO	D2
IR Sensor 1	OUT	D3
IR Sensor 2	OUT	D4
DHT11 Sensor	DATA	D5
ESP8266 TX	TX	D6 (RX)
ESP8266 RX	RX	D7 (TX via voltage divider)
Arduino 2 RX	RX	D8 (TX)
Arduino 2 TX	TX	D9 (RX)

Table 4.2 Arduino 2 (Slave Unit) Output Connections

Component	Pin	Arduino 2 Connection
LED	Anode	D8
Buzzer	Positive	D4
Servo Motor	Signal	D3
L298N IN1	Control	D5
L298N IN2	Control	D6
L298N ENA	PWM	D9
LCD (I2C)	SDA	A4
LCD (I2C)	SCL	A5

Table 4.3 4x4 Keypad Connections

Keypad Pin	Arduino 2 Pin
R1	D2
R2	D10
R3	D11
R4	D12
C1	D13
C2	A0
C3	A1
C4	A2

• Description of the Components

- LDR: Context-aware lighting based on ambient light levels.
- IR Sensors: Binary ”on/off” signals for decision making.
- Temperature Sensor: Controlling fans for comfort.
- Processing Units:
 - * Master: monitoring + rule evaluation.
 - * Slave: deterministic processing + user interface.

This enhances temporal efficiency and robustness

- Communication
 - * Local serial communication ensures fast coordination.
 - * Wi-Fi communication for remote operation.
- Maintain uniform logic levels.
- Prevent spurious operation due to VREF mismatch.

Table 4.4 Power Supply Requirements

Component	Voltage Requirement
Arduino Uno	5V
ESP8266	3.3V
L298N Motor Driver	12V
Sensors	5V
Servo Motor	5V

4.1.5 System Working and Operational Flow

The Smart Home Automation System operates by continuously monitoring environmental parameters and executing control actions based on predefined logic conditions. The system integrates sensing, decision-making, communication, and actuation in a coordinated manner to achieve intelligent automation. To better understand the working of the system, it is explained through different operating cases and scenarios.

Case-Based System Operation

Case 1: Automatic Lighting Control

- **Condition:** Presence detected and low ambient light
- **Observation:** The system identifies that a person is present in the room and the surrounding light intensity is below the threshold level.
- **Action:** The lighting system is automatically turned ON.
- **Condition:** No presence or sufficient ambient light
- **Observation:** Either no person is detected or the room already has adequate natural/artificial lighting.
- **Action:** The lighting system is turned OFF to conserve energy.

Case 2: Temperature-Based Fan Control

- **Condition:** Temperature below threshold
- **Action:** Fan remains OFF.
- **Condition:** Moderate temperature
- **Action:** Fan operates at low or medium speed.
- **Condition:** High temperature
- **Action:** Fan operates at maximum speed using PWM control.

This ensures comfort while optimizing power consumption.

Case 3: Occupancy Detection and Energy Saving

- **Condition:** Entry of a person into the room
- **Action:** System activates lighting and prepares for environmental control.
- **Condition:** Exit of all occupants
- **Action:** All appliances are automatically turned OFF.

This feature significantly reduces unnecessary energy usage.

Case 4: Security and Access Control

- **Condition:** Correct password entered via keypad
- **Action:** Door unlocking mechanism is activated.
- **Condition:** Incorrect password entered
- **Action:** Buzzer alert is triggered indicating unauthorized access.

This ensures basic security functionality in the system.

Case 5: Remote Control Operation

- **Condition:** User sends command through mobile application
- **Action:** System overrides automatic logic and executes user command.
- **Condition:** No remote command received
- **Action:** System continues to operate in automatic mode.

This provides flexibility between manual and automatic control.

Integrated System Behaviour

The overall system behavior is governed by the interaction of all subsystems. The sequence of operation is as follows:

1. The system continuously reads data from sensors. Environmental conditions such as light, temperature, and occupancy are evaluated.
2. Logical conditions are applied to determine required actions. The system checks for any remote user input.
3. Priority is given to user commands if present. Control signals are sent to output devices.
4. Output devices respond accordingly. The system updates status display and repeats the process.

System Characteristics

- Real-time response to environmental changes
- Automatic and manual control flexibility
- Energy-efficient operation
- Modular and scalable design

4.2 Codes and Standards

The design and implementation of the proposed Smart Home Automation System is based on established engineering principles, communication standards and software design methodology to ensure reliability, safety, interoperability and maintainability. These standards allow the system to behave consistently under different operating conditions and make it easier to modify and upgrade.

4.2.1 Codes and Standards

The implementation of the Smart Home Automation System follows relevant communication protocols, programming practices, and electrical safety guidelines to ensure reliable, efficient, and secure operation.

- **IEEE 802.11:** Standard for Wireless Local Area Network (WLAN) communication. It enables Wi-Fi connectivity for remote monitoring and control of the system.
- **UART Communication Standard:** Universal Asynchronous Receiver Transmitter (UART) is used for serial data communication between controllers and communication modules. It ensures stable and structured data transfer.
- **I2C Protocol (Inter-Integrated Circuit):** A two-wire communication protocol used for interfacing peripheral devices. It reduces wiring complexity and allows multiple devices to communicate efficiently.
- **PWM (Pulse Width Modulation):** A standard technique used for controlling power delivered to devices. It is used for regulating actuator operations such as fan speed control.
- **TCP/IP Protocol:** A standard communication protocol used for transmitting data over the internet. It enables interaction between the system and remote user interfaces.

- **Embedded C Programming Standards:** Structured and modular programming practices are followed to ensure code readability, maintainability, and efficient execution.
- **IEC 60364:** International standard for low-voltage electrical installations. Basic safety guidelines such as proper grounding, insulation, and voltage handling are considered.
- **General Electrical Safety Practices:** Standard precautions such as current limiting, proper insulation, and secure connections are followed to prevent component damage and ensure safe operation.

4.3 Constraints, Alternatives and Tradeoffs

4.3.1 Constraints

There are several constraints involved in the design of the Smart Home Automation System. Understanding these constraints helps in selecting an efficient and practical design strategy.

- **Hardware Constraints:**
 - Microcontrollers have limited processing power and memory.
 - This restricts:
 - * Complexity of algorithms.
 - * Number of simultaneous operations.
 - Therefore, the system uses simple logic and minimal data processing.
- **Sensor Limitations:**
 - Low-cost sensors may have:
 - * Limited accuracy.
 - * Environmental sensitivity.
 - To address this:
 - * Threshold-based logic is used.
 - * Minor fluctuations are ignored.

- **Communication Constraints:**

- Wireless communication introduces:
 - * Network availability issues.
 - * Signal interference.
 - * Delay in command transmission.
- The system is designed to operate offline when required.

- **Power Constraints:**

- Different components require different voltage levels.
- Improper power management may lead to:
 - * System instability.
 - * Component malfunction.
- Hence, proper power distribution is implemented.

4.3.2 Alternatives

Various design alternatives were considered before finalizing the system design.

- **Microcontroller Choice:**

- Alternative: High-performance computing platforms.
- Not selected due to:
 - * Higher cost.
 - * Increased complexity.
- Chosen: Cost-effective microcontroller.

- **Communication Technology:**

- Alternatives:
 - * Bluetooth (short range).
 - * GSM (higher cost).
- Chosen: Wi-Fi for better range and IoT compatibility.

- **System Architecture:**

- Alternative: Single controller system.
- Chosen: Dual controller system for better task distribution.

- **Control Mechanism:**

- Alternative: Direct ON/OFF control.
- Chosen: Driver-based control for improved performance.

4.3.3 Trade-offs

The design of the system involves balancing multiple engineering factors.

- **Cost vs Performance:**

- Low-cost components are used.
- Results in moderate performance and accuracy.

- **Complexity vs Efficiency:**

- Dual controller increases complexity.
- Improves efficiency and responsiveness.

- **Flexibility vs Reliability:**

- Wireless communication increases flexibility.
- Depends on network stability.

- **Accuracy vs Affordability:**

- Affordable sensors are used.
- Provides acceptable but not high precision.

CHAPTER 5

SCHEDULE, TASKS AND MILESTONES

5.1 Schedule

Table 5.1 Week-wise Project Schedule (16 Weeks)

Week	Phase	Activity
Week 1	Planning	Problem identification and objective definition
Week 2	Planning	Requirement analysis and system understanding
Week 3	Research	Literature review of existing systems
Week 4	Research	Study of IoT and automation technologies
Week 5	Design	System architecture design
Week 6	Design	Component selection (Arduino, sensors, ESP8266)
Week 7	Design	Circuit design and connection planning
Week 8	Design	Design validation and refinement
Week 9	Implementation	Hardware setup and wiring
Week 10	Implementation	Programming Arduino 1 (sensor logic)
Week 11	Implementation	Programming Arduino 2 (output control)
Week 12	Implementation	Communication setup between controllers
Week 13	Implementation	ESP8266 integration (IoT functionality)
Week 14	Integration	System integration and synchronization
Week 15	Testing	Testing, debugging, and performance evaluation
Week 16	Finalization	Documentation and final report preparation

5.2 Tasks and Milestones

In this section, the complete description of the tasks performed during the project and the milestones achieved is presented. The project work is divided into logical tasks, and the progress is evaluated through internal review stages conducted during the semester. These reviews ensure systematic development and proper validation of each phase of the project.

5.2.1 Tasks

The development of the Smart Home Automation System involves a series of structured tasks carried out in different phases.

- **Planning Tasks:**

- Identification of the problem statement.
- Definition of project objectives.
- Requirement analysis of the system.

- **Research Tasks:**

- Study of existing smart home automation systems.
- Analysis of IoT-based technologies.
- Understanding communication protocols.

- **Design Tasks:**

- Development of system architecture.
- Selection of hardware components.
- Design of circuit connections.

- **Implementation Tasks:**

- Hardware assembly and wiring.
- Programming Arduino 1 for sensor processing.
- Programming Arduino 2 for output control.
- Establishing communication between controllers.

- **Integration Tasks:**

- Integration of ESP8266 for IoT functionality.
- Synchronization of system modules.
- Ensuring proper data flow between components.

- **Testing Tasks:**

- Verification of sensor readings.
- Testing of output devices.
- Debugging and error correction.
- Performance evaluation of the system.

- **Documentation Tasks:**

- Preparation of thesis report.
- Creation of tables, diagrams, and figures.
- Final formatting and submission.

5.2.2 Milestones and Review Evaluation

The project progress is evaluated through internal reviews conducted during the semester.

- **Review 1:**

- Completion of planning and research tasks.
- Finalization of system design and architecture.
- Verification of feasibility of the proposed system.

- **Review 2:**

- Progress in hardware implementation.
- Development of software modules.
- Initial integration of system components.
- Partial testing and validation of system functionality.

Table 5.2 Project Milestones

Milestone	Description	Status
M1	Completion of Planning, Research and Design	Review 1 Completed.
M2	Partial Implementation and Integration Achieved	Review 2 Completed.
M3	Full System Integration Completed	Achieved.
M4	Testing and Validation Completed	Achieved.
M5	Final Documentation Completed	Achieved.

CHAPTER 6

PROJECT DEMONSTRATION

Figure 6.1 illustrates the complete hardware implementation of the Smart Home Automation System. The system is designed as a system of modules such as sensors, microcontrollers, communication modules, and output devices, which are interconnected. The temperature, light intensity and occupancy of an environment are monitored and processed in real time. On the basis of this data control actions are performed to provide work of appliances, including lighting and fan systems. The figure illustrates how the proposed automation system will work and be practically integrated.

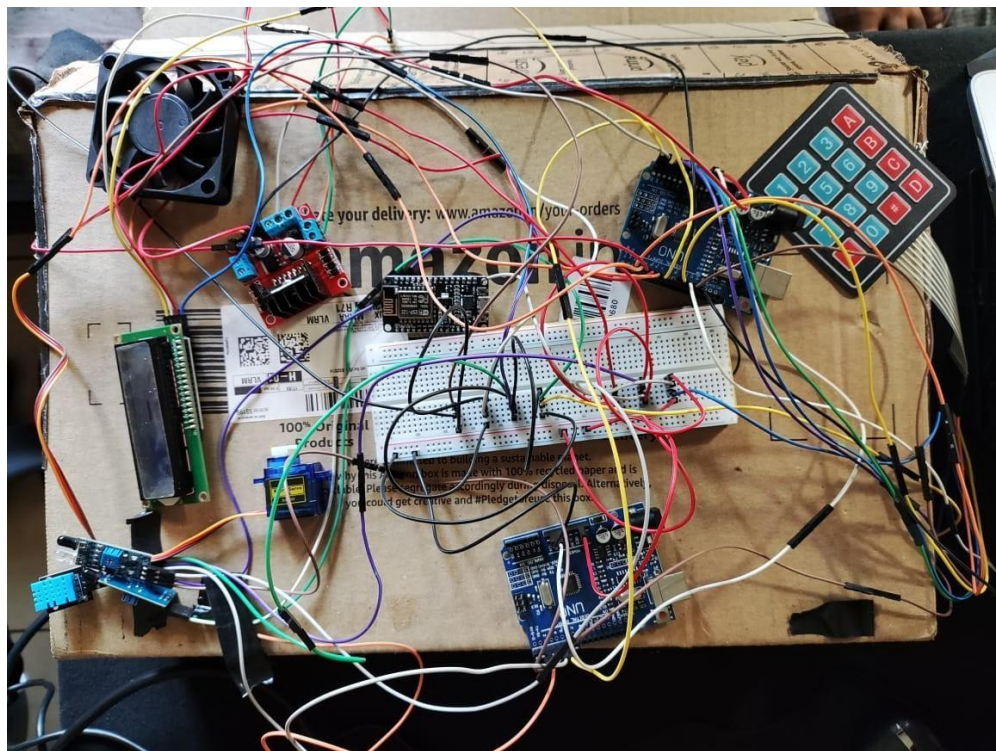


Figure 6.1 Smart Home Automation Hardware Setup

6.1 System Operation Under Occupancy

Figure 6.2 illustrates the Smart Home Automation System in active operation. The LCD module displays a temperature of 32.5°C, measured using the DHT11 temperature sensor, and detects the presence of two occupants through IR sensors. These inputs are processed by the Arduino microcontroller, which executes control logic in real time. Based on the sensed conditions, the system activates output devices such as the DC fan for cooling and controls lighting through the connected modules.

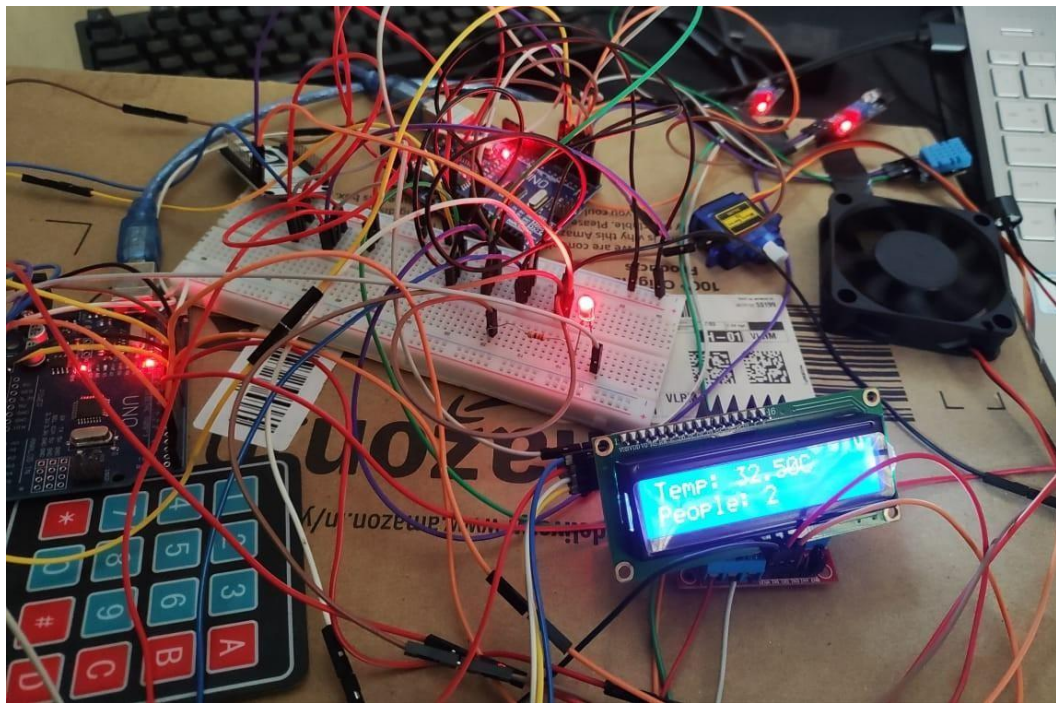


Figure 6.2 Real-Time System Output Display Showing Temperature of 32.5°C and Occupancy of 2 Persons

6.2 System Operation Under No Occupancy

Figure 6.3 illustrates the real-time operation of the Smart Home Automation System under no-occupancy conditions. The LCD module displays a temperature of 33.6°C, measured using the DHT11 temperature sensor, while zero occupants are detected through the IR sensors. These inputs are processed by the Arduino microcontroller, which executes the control logic accordingly. Based on the absence of occupants, the system automatically turns OFF output devices such as the DC fan and lighting modules to conserve energy.

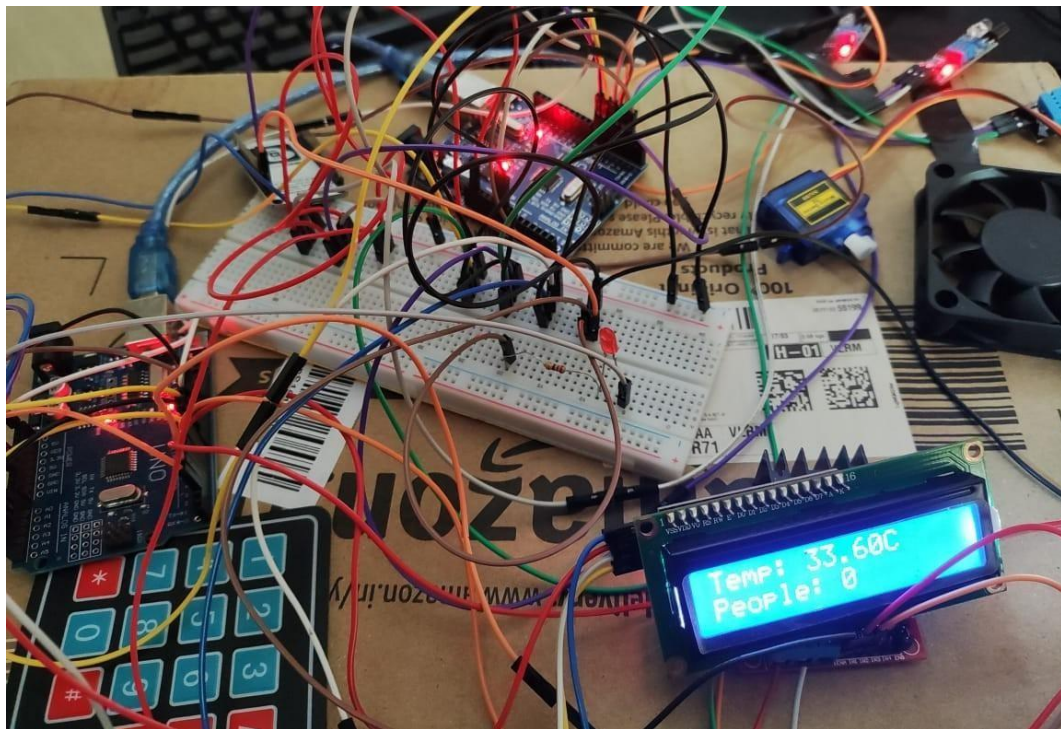


Figure 6.3 System Output Showing Temperature of 33.6°C and No Occupancy (All Appliances Turned OFF)

6.3 System Response to Valid Password

Figure 6.4 illustrates the operation of the security feature in the Smart Home Automation System. The LCD module displays the message “ACCESS OK”, indicating that the correct password has been entered through the 4×4 keypad. The keypad input is processed by the Arduino microcontroller, which verifies the entered password against the stored credentials. Upon successful authentication, the system activates the servo motor to unlock the door mechanism.

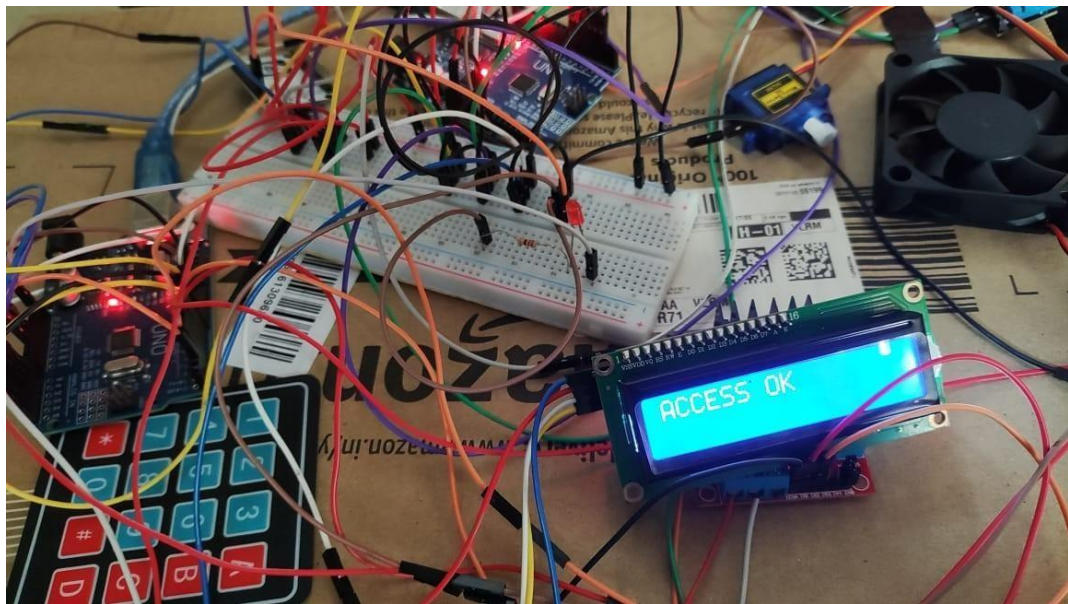


Figure 6.4 Security System Output Showing Successful Authentication (ACCESS OK)

6.4 System Response to Invalid Password

Figure 6.5 illustrates the operation of the security feature under incorrect password conditions in the Smart Home Automation System. The LCD module displays the message “WRONG PASS”, indicating that an invalid password has been entered through the 4×4 keypad. The keypad input is processed by the Arduino microcontroller, which compares it with the stored credentials. Upon detecting an incorrect match, the system denies access by keeping the servo motor in the locked position and activating buzzer.

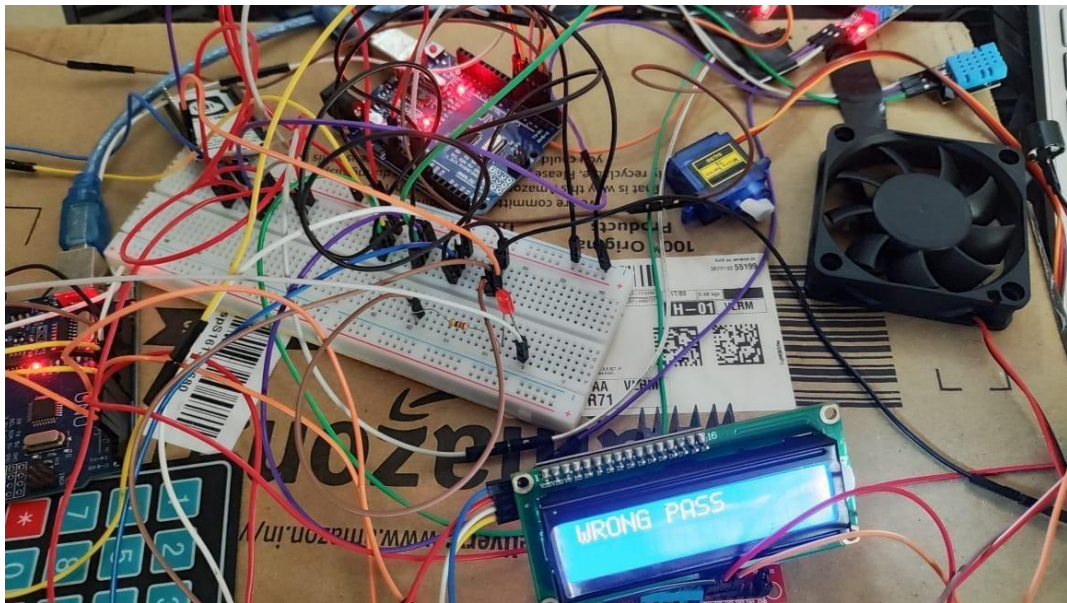


Figure 6.5 Security System Output Showing Incorrect Password Entry (WRONG PASS)

CHAPTER 7

COST ANALYSIS / RESULT AND DISCUSSION

7.1 Cost Analysis

The cost analysis of the Smart Home Automation System provides a detailed breakdown of all components used during the project development. The system is designed using cost-effective components while ensuring proper functionality and performance.

Table 7.1 Complete Component Cost Breakdown

S.No	Component	Qty	Unit Cost (Rs)	Total (Rs)
1	Arduino UNO R3 Board	2	324	648
2	ESP8266 NodeMCU	1	329	329
3	4-Channel Relay Module	1	249	249
4	DHT11 Temperature Sensor	1	199	199
5	LDR Sensor Module (LM393)	1	151	151
6	IR Sensors	2	89	178
7	DC Fan (12V)	1	130	130
8	L298N Motor Driver	1	191	191
9	4x4 Keypad Module	1	199	199
10	16x2 LCD Display (I2C)	1	226	226
11	Buzzer	1	70	70
12	Breadboard	1	158	158
13	Jumper Wires	2	133	266
14	Electronic Components Kit	1	399	399
15	12V Power Adapter	1	169	169
16	Connectors and Miscellaneous Components	1	95	95
Total				3657

7.1.1 Cost Analysis Discussion

The total cost of the system is approximately **Rs 3657**. The inclusion of additional components such as LCD display, IR sensors, and buzzer enhances the functionality of the system while maintaining affordability.

The use of two Arduino boards increases the cost but improves system performance by distributing tasks between sensing and output control. The ESP8266 module enables IoT-based control, making the system more advanced.

Most of the cost is attributed to core components such as microcontrollers and modules, while supporting components contribute moderately to the overall budget.

7.2 Results and Discussion

7.2.1 Key Findings

- The created system was able to successfully automate the process of lighting according to occupancy and ambient light conditions. It was noted that the light was kept at OFF position when there was adequate natural lighting and the lighting was set to automatically turn ON when the environment was dark and occupancy was detected. This illustrates good energy saving behavior when put under realistic situations.
- The fan control mechanism based on temperature worked well in various temperature conditions. PWM control was used to dynamically adjust the speed of the fan, ensuring that the fan provides a good level of thermal comfort while minimizing unnecessary power usage.
- The occupancy detection with the IR sensors used in monitoring the entrance and exit conditions was effectively used. The system also made sure that the appliances were switched off in the absence of any occupant, thus minimizing on the amount of energy wasted.
- The combination of the keypad-based security system allowed the control of the access to the system. Successful unlocking of doors occurred when correct passwords were typed in and failure occurred when incorrect passwords were typed in and a buzzer alarm sounded, which confirmed the security functionality.
- The two microcontrollers communicated effectively and efficiently. The independent handling between sensing and actuation activities led to a higher level of responsiveness and reduced processing delays on the part of the system. The LCD module was used to display system parameters including temperature, occupancy status and device states show up in real time and provide the user with continuous feedback.

- The remote control functionality of the IoT enabled users to monitor and manage appliances with the help of a mobile application. Response to user commands was effective and a demonstration of a successful implementation of remote automation.

7.2.2 Discussion

The findings prove that the Smart Home Automation System is effective in integrating sensing, processing, and control systems to attain intelligent automation. The system reacted properly to changes in the environment giving it a high degree of control over the appliances based on a predefined logic.

This was adequately enough to use the low-cost sensors towards the general automation tasks but it was observed that there were minor fluctuations as a result of the environmental variations. Nonetheless, threshold-based logic implementation guaranteed the stable and reliable operation despite such differences.

The dual microcontroller platform greatly improved the performance of the systems since duties were shared between sensing and output control. This minimized the computational effort on a per-controller basis, and enhanced real-time responsiveness.

The remote control made possible by wireless communication enhanced flexibility in the system. But it was noted that stability of the network is related to stability of system performance, and delays can occur in case of poor network connectivity.

On the whole, the system indicates a balance between the cost, performance, and functionality. Simple control logic and the use of low cost components makes the system applicable to real world applications whilst maintaining an acceptable level of efficiency and reliability.

7.2.3 Future Scope

Although the system performs effectively, several improvements can be considered for future development:

- Integration of advanced sensors for higher accuracy and improved reliability.
- Implementation of machine learning algorithms for predictive automation and intelligent decision-making.
- Development of a more advanced mobile application with enhanced user interface and control features.
- Integration with voice assistants such as Google Assistant or Alexa for hands-free operation.

- Expansion of the system to support additional smart devices and larger-scale automation.
- Enhancement of system security through encryption and advanced authentication mechanisms.

7.2.4 Results Based on System Demonstration

The experimental results obtained from the hardware implementation validate the successful operation of the Smart Home Automation System. The complete setup, as shown in Fig 6.1, consists of interconnected modules including Arduino microcontrollers, ESP8266 communication module, DHT11 temperature sensor, IR sensors, 4×4 keypad, LCD display, servo motor, DC fan, LEDs, and supporting circuitry on the breadboard.

Figure 6.2 demonstrates the system under active occupancy conditions. The DHT11 sensor measures a temperature of 32.5°C, while the IR sensors detect the presence of two occupants. These inputs are processed by the Arduino microcontroller, which executes the control logic in real time. Based on the sensed conditions, the system activates output devices such as the DC fan for cooling and controls lighting through connected modules. The LCD display provides real-time feedback, showing temperature and occupancy status, confirming proper integration of sensing, processing, and actuation units.

Figure 6.3 represents the system under no-occupancy conditions. The LCD module displays a temperature of 33.6°C with zero occupants detected by the IR sensors. The Arduino processes this input and automatically turns OFF output devices such as the fan and lighting system. The inactive state of LEDs and fan confirms the correct implementation of occupancy-based energy-saving logic, demonstrating efficient resource utilization.

Figures 6.4 and 6.5 illustrate the operation of the security subsystem. In Fig 6.4, the correct password entered through the 4×4 keypad is processed by the Arduino, and the LCD displays “ACCESS OK”. The system responds by activating the servo motor to unlock the door mechanism, confirming successful authentication. In contrast, Fig 6.5 shows the case of incorrect password entry, where the LCD displays “WRONG PASS”. The Arduino denies access by keeping the servo motor in the locked position and activating alert indicators such as LEDs or buzzer, ensuring system security.

Overall, the results confirm that the Smart Home Automation System effectively integrates multiple hardware components and control strategies. The system successfully performs real-time sensing using DHT11 and IR sensors, processes inputs using Arduino, communicates through ESP8266, and controls output devices such as fan, lighting, and door mechanism. The LCD module provides continuous feedback, while the keypad enables user interaction and security control. The coordinated operation of all modules demonstrates reliable performance, energy efficiency, and practical feasibility of the proposed system.

CHAPTER 8

SUSTAINABLE DEVELOPMENT GOALS (SDG) CONTRIBUTION

8.1 Introduction

The Sustainable Development Goals (SDGs) set by the United Nations provide an international consensus on how to tackle pressing issues such as energy efficiency, sustainable infrastructure, and resource management. Modern engineering projects aim to comply with these objectives to ensure a positive impact on society and the environment through technological development.

The Smart Home Automation System created in this project supports these goals by incorporating smart control strategies that enable energy efficiency, efficiency and sustainability. Through automation, real-time monitoring and control, the system helps to minimise energy waste and improve resource efficiency.

8.2 Relevant Sustainable Development Goals

The proposed system contributes to multiple SDGs as outlined below:

8.2.1 SDG 7: Affordable and Clean Energy

This project helps in achieving SDG 7, which is to ensure efficient energy use in buildings. The automation system ensures that appliances are only powered when needed, depending on various environmental factors such as occupancy, light levels and temperature. This eliminates wastage of electricity and promotes energy savings.

Moreover, the low-power embedded systems and control logic design improves energy efficiency, making the solution cost-effective and environmentally friendly.

8.2.2 SDG 9: Industry, Innovation and Infrastructure

This project helps in achieving SDG 7, which is to ensure efficient energy use in buildings. The automation system ensures that appliances are only powered when needed, depending on various environmental factors such as occupancy, light levels and temperature. This eliminates wastage of electricity and promotes energy savings.

Moreover, the low-power embedded systems and control logic design improves energy efficiency, making the solution cost-effective and environmentally friendly.

8.2.3 SDG 11: Sustainable Cities and Communities

Smart homes are essential in the creation of sustainable cities. This project helps achieve SDG 11 by facilitating the efficient use of household resources and enhancing living conditions through automation.

Automatic lighting, temperature control, and monitoring functionalities eliminate unnecessary energy consumption and improve ease of use, contributing to a sustainable and smart home.

8.2.4 SDG 12: Responsible Consumption and Production

The system enables sustainable consumption patterns by efficiently using the energy resource and when needed. Automated controls prevent unnecessary usage, thus avoiding energy wastage.

The accessibility of the system through the use of affordable and easily accessible materials also promotes sustainable manufacturing, ensuring that it can be easily implemented.

CHAPTER 9

SUMMARY

This project focused on the design and implementation of a Smart Home Automation System using Arduino and IoT technology. The primary objective was to develop a system capable of automatically controlling household appliances based on environmental conditions while also allowing remote user interaction. The system was expected to provide efficient automation, ease of use, and cost-effectiveness.

The results obtained from the project demonstrate that the system successfully meets these expectations. The sensors effectively monitor parameters such as light intensity, temperature, and occupancy, enabling automatic control of lighting and fan systems. The integration of the ESP8266 module allows remote access through a mobile application, enhancing user convenience. Additionally, the use of a dual microcontroller architecture improves system performance by distributing tasks between sensing and control operations. Overall, the system performs reliably and achieves the intended objectives of automation, flexibility, and affordability.

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Capstone Project

Project Title : Smart Home Automation System using Arduino
Team Members : PARAS SINGH BISHT (22BEE0331)
SNEHIL RAJ 2 (22BEE0079)
Faculty Guide : Dr.Sathish Kumar K, Proffesor Grade 2
School Of Electrical Engineering /VIT Vellore
Semester/Year : VIII Semester, May 2026

Project Abstract:

The aim of this project is to develop a Smart Home Automation System to enhance energy efficiency, comfort and security. Sensors are used to measure various environmental factors including light, temperature and occupancy. Using these data, decisions are made to operate appliances such as lights and fans.

It is implemented with a distributed architecture in which sensing and control tasks are performed separately for better performance and reliability. It also supports remote access, enabling users to access and control devices using a mobile app. Other features, including password protection and real-time display, improve the system's user experience. The research has shown an affordable and scalable solution to home automation.

Codes and Standards:

IEEE 802.11 (Wi-Fi Communication Standard); UART Communication Protocol; I2C Communication Protocol; PWM Control Technique; TCP/IP Protocol; IEC 60364 (Electrical Safety Standard).

Significant Design Constraints:

1. **Processing and Memory Constraints:** The microcontroller has limited compute power and memory, limiting the complexity of algorithms and the number of simultaneous operations.

2. **Sensor Limitations:** The sensors used in the system may not be highly accuracy and may be susceptible to environmental factors, impacting measurement accuracy.
3. **Communication Constraints:** Wireless networking is subject to network connectivity and can be subject to lags or interference, impacting real-time control.
4. **Power Constraints:** Components operate at different voltage levels, so it is making good power management crucial for system stability.

Significant Tradeoffs:

1. *Cost vs. Performance:* Affordable components are selected to reduce the cost able, which may come at the expense of some accuracy and speed. systems.
2. *Complexity vs. Efficiency:* A dual microcontroller system increases system but enhances performance and multitasking.
3. *Flexibility vs. Reliability:* Wireless control is flexible but requires stable network connectivity.

Computing Aspects:

Data is acquired and controlled using embedded programming. Data from sensors is analysed for logical decision-making in automation. Communication between different modules is handled using serial protocols, while remote communication is provided by network interfaces. The system operates in a loop to provide real-time interaction and control of devices.

CHAPTER A

Codes

```
1 #include <Servo.h>
2 #include <LiquidCrystal_I2C.h>
3 #include <Keypad.h>
4 #include <SoftwareSerial.h>
5 SoftwareSerial master(7,6);
6 LiquidCrystal_I2C lcd(0x27,16,2);
7 Servo door;
8 #define LED 8
9 #define IN1 5
10 #define IN2 6
11 #define ENA 9
12 #define BUZZ 4
13 const byte ROWS=4, COLS=4;
14 char keys[ROWS][COLS]={
15   {'1','2','3','A'},
16   {'4','5','6','B'},
17   {'7','8','9','C'},
18   {'*','0','#','D'}
19 };
20 byte rowPins[ROWS]={2,10,11,12};
21 byte colPins[COLS]={13,A0,A1,A2};
22 Keypad keypad=Keypad(makeKeymap(keys),rowPins,colPins,ROWS,COLS
    );4
23 String pass="1234", input="";
24 void setup(){
25   master.begin(9600);
26   pinMode(LED,OUTPUT);
27   pinMode(IN1,OUTPUT);
```

Listing A.1: Arduino SLAVE Controller Logic


```

1
2  pinMode(IN2,OUTPUT);
3  pinMode(ENA,OUTPUT);
4  pinMode(BUZZ,OUTPUT);
5  door.attach(3);
6  lcd.init();
7  lcd.backlight();
8  }
9  void loop() {
10     if(master.available()){
11         String cmd=master.readStringUntil('\n');
12         if(cmd=="L1") digitalWrite(LED,HIGH);
13         if(cmd=="L0") digitalWrite(LED,LOW);
14         if(cmd=="F0") analogWrite(ENA,0);
15         if(cmd=="F1") analogWrite(ENA,120);
16         if(cmd=="F2") analogWrite(ENA,180);
17         if(cmd=="F3") analogWrite(ENA,255);
18         if(cmd=="D1") door.write(90);
19         if(cmd=="D0") door.write(0);
20     }
21     char key=keypad.getKey();
22     if(key) {
23         input+=key;
24         if(input.length()==4) {
25             if(input==pass) {
26                 door.write(90);
27                 lcd.clear(); lcd.print("ACCESS OK");
28             } else {
29                 digitalWrite(BUZZ,HIGH);
30                 delay(1000);
31                 digitalWrite(BUZZ,LOW);
32                 lcd.clear(); lcd.print("WRONG PASS");
33             }
34             input="";
35         }
36     }
37 }

```

Listing A.2: Arduino SLAVE Controller Logic

```

1  #include <SoftwareSerial.h>
2  #include <DHT.h>
3  #define DHTPIN 5
4  #define DHTTYPE DHT11
5  SoftwareSerial esp(6,7);
6  SoftwareSerial slave(9,8);
7  DHT dht(DHTPIN, DHTTYPE);
8  int people = 0;
9  void setup() {
10     pinMode(2, INPUT);
11     pinMode(3, INPUT);
12     pinMode(4, INPUT);
13     esp.begin(9600);
14     slave.begin(9600);
15     dht.begin();
16 }
17 void loop() {
18     int light = digitalRead(2);
19     int ir1 = digitalRead(3);
20     int ir2 = digitalRead(4);
21     float temp = dht.readTemperature();
22     if(ir1==0) people++;
23     if(ir2==0 && people>0) people--;
24     if(people>0 && light==0) slave.println("L1");
25     else slave.println("L0");
26     if(people==0) slave.println("F0");
27     else if(temp<25) slave.println("F0");
28     else if(temp<30) slave.println("F1");
29     else if(temp<35) slave.println("F2");
30     else slave.println("F3");
31     if(esp.available()) {
32         String cmd = esp.readStringUntil('\n');
33         slave.println(cmd);
34     }
35     delay(1000);
36 }

```

Listing A.3: Arduino MASTER Controller Logic

```

1  #include <ESP8266WiFi.h>
2
3  void setup() {
4      Serial.begin(9600);
5      WiFi.begin("SSID", "PASSWORD"); // \ac{IoT} connection
6  }
7
8  void loop() {
9      if (Serial.available()) {
10         String cmd = Serial.readString();
11         // Logic to send data to Mobile App
12     }
13 }#define BLYNK_PRINT Serial
14 #include <ESP8266WiFi.h>
15 #include <BlynkSimpleEsp8266.h>
16
17 char auth[] = "YOUR_AUTH";
18 char ssid[] = "WIFI";
19 char pass[] = "PASS";
20
21 void sendCmd(String c){ Serial.println(c); }
22
23 BLYNK_WRITE(V0){ if(param.asInt()) sendCmd("L1"); else sendCmd(
    "L0"); }
24 BLYNK_WRITE(V1){ if(param.asInt()) sendCmd("F1"); }
25 BLYNK_WRITE(V2){ if(param.asInt()) sendCmd("F2"); }
26 BLYNK_WRITE(V3){ if(param.asInt()) sendCmd("D1"); }
27 BLYNK_WRITE(V4){ if(param.asInt()) sendCmd("D0"); }
28
29 void setup(){
30     Serial.begin(9600);
31     Blynk.begin(auth,ssid,pass);
32 }
33
34 void loop(){
35     Blynk.run();
36 }

```

Listing A.4: ESP8266 IoT